

UNIT I: PHYSICAL LAYER, DELAYS & DATA LINK PROTOCOLS

Weightage: 25% | Syllabus Section 1

1. Transmission Delays & Bandwidth Limits

Transmission Delay: $T_t = L/R$ (L packet bits, R bps bandwidth).

Propagation Delay: $T_p = d/v$ (d distance, $v \approx 2 \times 10^8$ m/s).

Nyquist Bit Rate (Noiseless): $C = 2B \log_2 M$ bps.

Shannon Capacity (Noisy): $C = B \log_2(1 + \text{SNR})$ bps, where $\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR})$.

2. Sliding Window Protocol Efficiency

Parameter a: $a = T_p/T_t$. Round Trip Time $\text{RTT} = 2T_p$.

Stop-and-Wait: Efficiency $\eta = \frac{1}{1+2a}$.

Go-Back-N (GBN): Window size $N \leq 2^k - 1$. Efficiency: $\eta = \min\left(1, \frac{N}{1+2a}\right)$. Receiver window = 1.

Selective Repeat (SR): Sender & Receiver windows $W_s = W_r = 2^{k-1}$. Independent ACKs; retransmits only damaged frames.

3. Error Detection & CRC Division

CRC Polynomial: Data $D(x)$ appended with r zeros divided by generator $G(x)$ of degree r using XOR arithmetic.

- Transmitted Code: $T(x) = D(x) \cdot 2^r \oplus R(x)$.
- Catches all single-bit errors, burst errors of length $\leq r$.

UNIT II: MAC SUBLAYER & ETHERNET CSMA/CD

Weightage: 25% | Syllabus Section 2

1. ALOHA & CSMA/CD Collisions

Pure ALOHA: Max throughput $S = Ge^{-2G} \implies S_{\text{max}} = \frac{1}{2e} \approx 18.4\%$ at $G = 0.5$. Vulnerable time = $2T_{fr}$.

Slotted ALOHA: Max throughput $S = Ge^{-G} \implies S_{\text{max}} = \frac{1}{e} \approx 36.8\%$ at $G = 1.0$.

CSMA/CD Minimum Frame Size: Must transmit long enough to detect collision across furthest node:

$$T_t \geq 2T_p \implies \frac{L_{\text{min}}}{R} \geq 2\frac{d}{v} \implies L_{\text{min}} = 2 \cdot R \cdot T_p.$$

2. Exponential Backoff Algorithm

Collision Backoff: After i collisions, choose random slots $k \in [0, 2^{\min(i,10)} - 1]$. Wait time: $k \times 2T_p$. Abort if collisions reach 16!

UNIT III: NETWORK LAYER, IPV4 SUBNETTING & ROUTING

Weightage: 25% | Syllabus Section 3

1. CIDR Subnetting Calculations

IPv4 Address: 32-bit address. In CIDR /n notation, network mask has n ones, $32 - n$ zeros.

Formulas:

- Total IP addresses in subnet: 2^{32-n} .
- Usable host addresses: $2^{32-n} - 2$ (Network ID and Directed Broadcast address reserved!).

2. Distance Vector Routing (DVR)

Bellman-Ford Equation: $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$, where v spans neighbors of x .

Count-to-Infinity Problem: Caused by routing loops on link failure. Solved by: (1) Split Horizon, (2) Poison Reverse.

3. Link State Routing (LSR)

Dijkstra's Algorithm: Each node floods Link State Packets (LSP) to discover complete network topology map, then independently executes Dijkstra to construct shortest path tree!

UNIT IV: TRANSPORT LAYER (TCP/UDP) & CONGESTION CONTROL

Weightage: 25% | Syllabus Section 4

1. TCP Congestion Control State Machine

Slow Start: Initial $CWND = 1$ MSS. Grows exponentially: doubles every RTT ($CWND = 2 \times CWND$) until reaching $ssthresh$.

Congestion Avoidance: Linear additive increase: $CWND = CWND + 1$ MSS per RTT.

On Timeout (Severe): Multiplicative decrease: $ssthresh = \lfloor CWND/2 \rfloor$; reset $CWND = 1$ MSS.

On 3 Duplicate ACKs (Mild): Fast Retransmit + Fast Recovery: $ssthresh = \lfloor CWND/2 \rfloor$, set $CWND = ssthresh + 3$.

2. TCP Flow Control & Windowing

Sender Window: $W = \min(CWND, RWND)$, dynamically balancing network congestion and receiver buffer capacity.

Silly Window Syndrome: Prevented by Nagle's algorithm at sender, Clark's solution at receiver.